USB Camera Introduction

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1. Content Description

- Start the camera driver, view the camera image through rqt, and check the camera intrinsic parameters topic

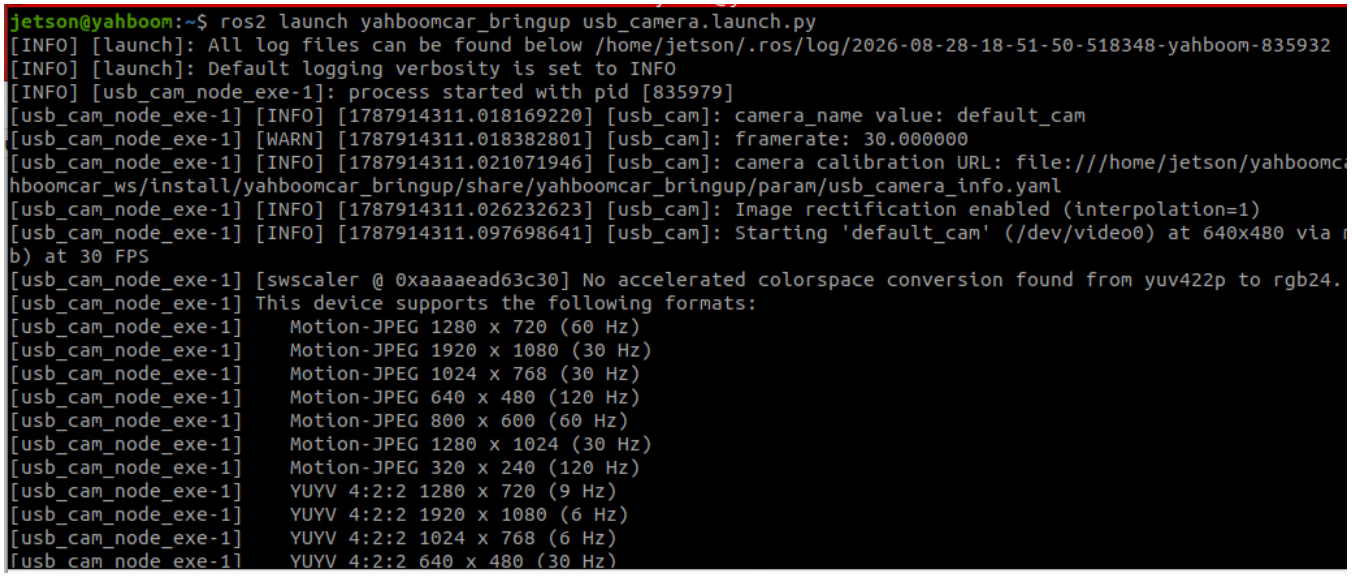
2. Camera Introduction



Lens solution	CMOS
Image sensor	1/2.7inch OV2710
Lens	F2.8mm
Image pixels	2 MP
Image format	YUV/MJPG
Resolution	240p/480p/720p/1080p
Max frame rate	120fps
FOV (diagonal)	148°
Distortion	-17%
USB	USB2.0 driver-free
Camera size	32*32*29.5mm

3. Preparation

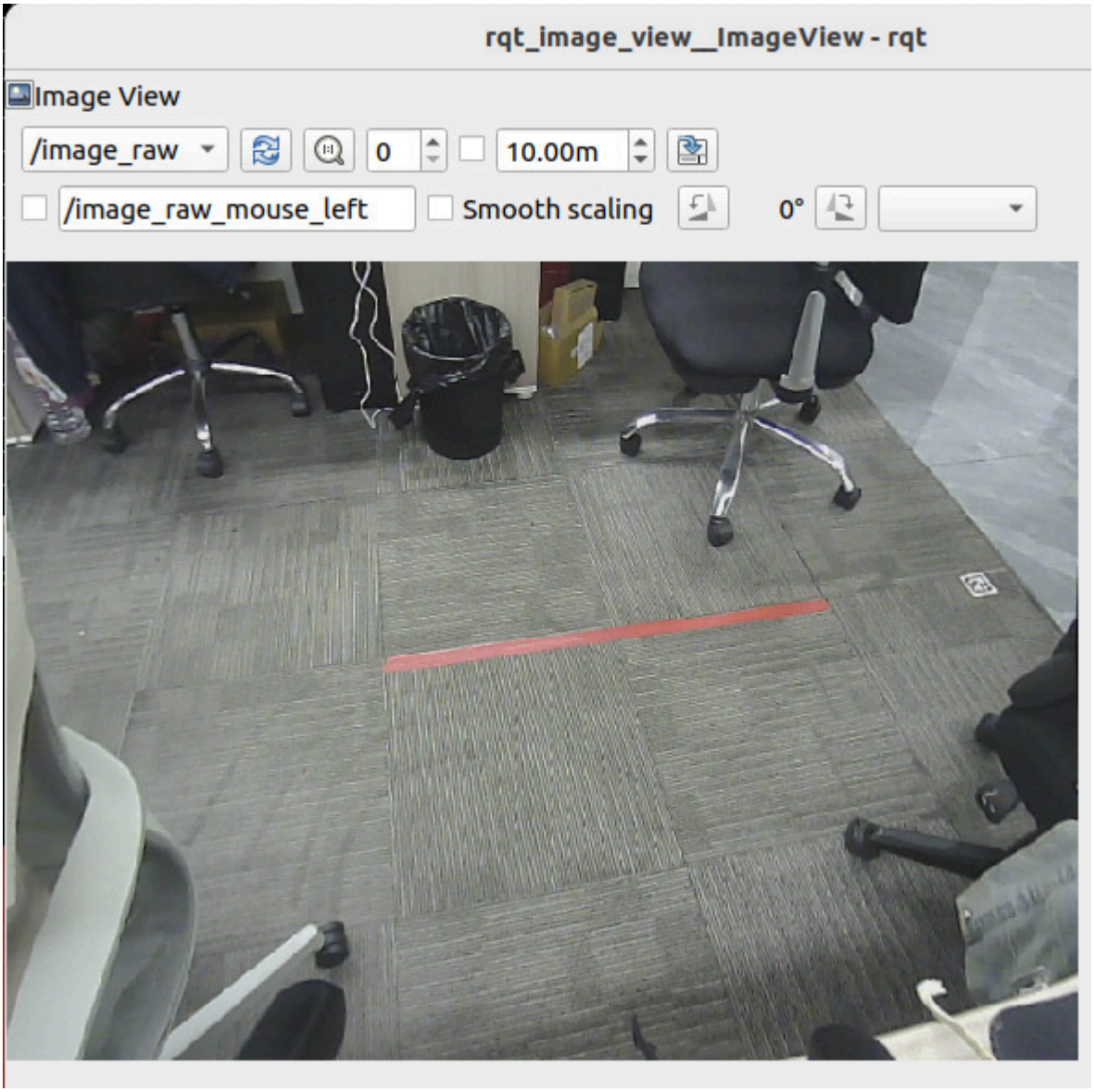
```
ros2 launch yahboomcar_bringup usb_camera.launch.py
```



4. Case Demonstration

```
ros2 run rqt_image_view rqt_image_view
```

- Select the `/image_raw` topic to see the camera's RGB color image



- Check the camera intrinsic parameters topic

```
ros2 topic echo /camera_info --once
```

- The camera's intrinsic parameters topic will be used in subsequent courses for functions such as Aruco marker pose solving and PnP pose solving

```
jetson@yahboom:~$ ros2 topic echo /camera_info --once
header:
  stamp:
    sec: 1787914680
    nanosec: 884997000
  frame_id: camera
height: 480
width: 640
distortion_model: plumb_bob
d:
- 0.0
- 0.0
- 0.0
- 0.0
- 0.0
k:
- 401.24164
- 0.0
- 316.34302
- 0.0
- 445.96533
- 230.08756
- 0.0
- 0.0
- 1.0
```

5. Source Code Analysis

- Launch path

```
~/yahboomcar_ros2_ws/yahboomcar_ws/src/yahboomcar_bringup/launch/usb_camera.launch.py
```

- Parameter file

```
~/yahboomcar_ros2_ws/yahboomcar_ws/src/yahboomcar_bringup/param/common.yaml
```

- `common_params`: loads the `common.yaml` configuration file through `FilePathManager`, including parameters such as the camera IO method and pixel format
- `camera_info_file`: concatenates the full path of the camera intrinsic parameters file

`usb_camera_info.yaml`, adding the `file://` prefix for the driver to load

- `uasb_camera`: starts the `usb_cam_node_exe` node, with the main parameters:
    - `video_device`: video device path `/dev/video0`
    - `image_width` / `image_height`: resolution 640×480
    - `framerate`: frame rate 30fps
    - `camera_info_url`: path of the intrinsic parameters file, used to publish the `/camera_info` topic
    - `enable_rectify`: enable image rectification

```
def generate_launch_description():
    ld = LaunchDescription()
    common_params =  FilePathManager().GetFileAndCheckExist("config_file")
    camera_info_file = 'file://' +
os.path.join(get_package_share_directory('yahboomcar_bringup'),'param','usb_camera_info.yaml
')
    uasb_camera = Node(
        package='usb_cam',
        executable='usb_cam_node_exe',
        name='usb_cam',
        output='both',
        sigterm_timeout='15',
        sigkill_timeout='15',
        parameters=[common_params,
            {'video_device': '/dev/video0'},
            {'image_width': 640},
            {'image_height': 480},
            {'framerate': 30.0},
            {'camera_info_url': camera_info_file},
            {'enable_rectify': True}
        ],
        arguments=['--ros-args', '--log-level', LaunchConfiguration('log_level')],
    )

    ld.add_action(DeclareLaunchArgument('log_level', default_value='info',description='ros2
logging level'))
    ld.add_action(uasb_camera)
    return ld
```

- `io_method`: image acquisition method; `mmap` is memory mapping, which is more efficient
- `frame_id`: name of the camera TF coordinate frame, here `camera`
- `pixel_format`: pixel format conversion; `yuyv2rgb` converts the raw YUYV format to RGB
- `av_device_format`: raw device data format `YUV422P`
- `brightness` / `contrast` / `saturation` / `sharpness` / `gain`: brightness, contrast, saturation, sharpness, gain; `-1` means use the camera default value
- `auto_white_balance`: enable automatic white balance
- `white_balance`: white balance color temperature 6000K (takes effect when automatic white balance is

off)

- `autoexposure`: enable automatic exposure
- `exposure`: exposure value 80 (takes effect when automatic exposure is off)
- `focus`: focus value; `-1` means use the camera default focus

> **Tip**
>
> - If the ambient light changes a lot during use, causing the image to be overexposed or too dark, you can turn off **autoexposure** and manually adjust the **exposure** value

```
usb_cam:
    ros__parameters:
      io_method: "mmap"
      frame_id: "camera"
      pixel_format: "yuyv2rgb"
      av_device_format: "YUV422P"
      brightness: -1
      contrast: -1
      saturation: -1
      sharpness: -1
      gain: -1
      auto_white_balance: true
      white_balance: 6000
      autoexposure: true
      exposure: 80
      focus: -1
```

- Adjusting the exposure value of the USB monocular camera: as shown in the figure below, if the image is overexposed or too dark, you can turn off the camera's automatic exposure and manually adjust `exposure`

